

HSINCHU CITY, Taiwan

WMO: 467570

Lat: 24.8278N

Lon: 121.0142E

Elev: 27

StdP: 101.00

Time Zone: 8.00 (E08)

Period: 90-98

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.1	10.4	3.7	5.0	10.9	5.3	5.5	12.3	10.8	13.8	9.9	13.6	4.5	50	0.342

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.3	33.2	26.7	32.5	26.5	31.9	26.4	27.6	31.6	27.2	31.3	26.8	30.9	4.0	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.5	22.1	30.1	26.0	21.5	29.8	25.6	20.9	29.6	88.1	31.6	86.2	31.6	84.5	31.0	29.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.3	9.2	8.2	5.1	36.0	3.2	2.2	2.8	37.6	0.9	38.9	-0.9	40.2	-3.2	41.8		
			4.4	28.5	2.7	0.7	2.5	29.0	0.9	29.4	-0.6	29.8	-2.6	30.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	22.6	15.5	15.7	18.0	21.6	24.8	27.3	28.8	28.6	26.7	23.8	21.5	18.2
	DBStd	5.22	2.68	2.78	3.13	3.34	2.56	2.09	1.13	1.13	1.84	2.20	2.41	2.54
	HDD10.0	2	2	0	0	0	0	0	0	0	0	0	0	1
	HDD18.3	267	91	83	45	11	1	0	0	0	0	0	4	32
	CDD10.0	4590	173	159	247	349	457	518	582	576	502	427	344	256
	CDD18.3	1812	4	9	34	110	199	268	323	317	252	169	98	28
	CDH23.3	16602	4	14	94	571	1563	2937	4014	3739	2396	900	331	39
CDH26.7	5669	0	0	9	84	403	1083	1714	1453	726	153	41	2	
Wind	WSAvg	3.2	3.6	3.4	2.9	2.5	2.1	2.7	2.5	2.4	3.1	4.4	4.1	4.3
Precipitation	PrecAvg	2403	128	151	139	149	231	242	190	249	330	281	205	137
	PrecMax	3931	311	364	232	459	520	474	412	585	1112	1179	520	332
	PrecMin	1293	45	27	49	27	56	75	32	61	61	40	34	14
	PrecStd	584	71	88	41	94	121	100	111	153	229	232	135	86
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.8	25.2	27.9	30.1	32.7	33.0	34.2	34.2	33.5	30.9	29.5	26.2
		MCWB	18.0	20.1	22.7	24.1	26.4	26.4	26.6	26.7	26.2	24.8	23.6	21.1
	2%	DB	21.9	23.2	25.7	28.7	31.2	32.1	33.2	33.0	32.2	29.5	27.9	24.3
		MCWB	17.3	19.0	21.5	23.8	25.8	26.6	26.8	26.8	25.9	24.2	22.2	20.1
	5%	DB	20.6	21.6	24.1	27.6	29.9	31.5	32.5	32.3	31.2	28.4	26.6	22.9
		MCWB	16.8	17.9	20.5	23.4	25.2	26.3	26.6	26.6	25.6	23.5	21.7	19.2
	10%	DB	19.2	19.9	22.7	26.5	28.8	30.9	31.8	31.6	30.2	27.3	25.3	21.9
		MCWB	16.2	16.9	19.8	23.0	24.6	26.0	26.4	26.4	25.3	22.9	21.1	18.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.1	21.3	23.2	25.1	27.2	28.3	28.1	28.2	27.2	26.1	24.0	22.3
		MCDB	21.9	24.0	26.1	28.8	31.5	31.1	32.4	32.2	31.8	29.7	27.7	25.1
	2%	WB	18.2	19.5	22.0	24.4	26.2	27.2	27.6	27.5	26.7	25.2	23.3	21.1
		MCDB	20.8	22.6	25.1	27.7	30.1	30.8	31.9	31.6	30.9	28.1	26.6	23.3
	5%	WB	17.3	18.3	21.1	23.8	25.5	26.7	27.1	27.1	26.2	24.5	22.5	20.1
		MCDB	19.7	20.9	23.8	27.0	29.1	30.5	31.4	31.2	30.1	27.3	25.7	22.3
	10%	WB	16.6	17.2	20.1	23.1	25.0	26.2	26.7	26.7	25.7	23.7	21.7	19.1
		MCDB	18.7	19.4	22.1	26.2	28.4	30.0	30.9	30.6	29.2	26.5	24.7	21.4
Mean Daily Temperature Range	5% DB	MDBR	5.3	5.0	5.3	5.6	5.9	5.6	6.3	6.0	6.1	5.5	6.0	5.4
		MCDBR	7.9	8.1	8.0	7.2	7.1	6.4	6.9	7.1	7.5	7.2	8.2	7.5
		MCWBR	4.8	4.8	4.4	3.8	3.8	3.5	3.4	3.2	3.6	4.0	4.2	4.5
	5% WB	MCDBR	6.7	7.2	7.4	6.5	6.7	6.1	6.6	6.5	6.7	6.0	7.3	6.2
		MCWBR	4.4	4.5	4.3	3.8	3.9	3.5	3.5	3.4	3.4	3.6	4.3	4.1
Clear-Sky Solar Irradiance	taub	0.542	0.588	0.701	0.706	0.603	0.559	0.552	0.574	0.554	0.541	0.532	0.538	
	taud	1.880	1.802	1.598	1.638	1.900	2.032	2.046	1.979	2.012	1.991	1.988	1.906	
	Ebn at Noon	699	700	647	659	730	757	761	745	749	733	705	681	
	Edh at Noon	179	206	265	260	199	173	171	182	173	168	159	168	
All-Sky Solar Radiation	RadAvg	2.43	2.72	3.31	3.71	4.31	4.93	5.96	5.41	4.71	3.88	2.79	2.40	
	RadStd	0.44	0.49	0.43	0.54	0.74	0.58	0.35	0.50	0.49	0.43	0.33	0.37	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (20 neighbors)	+0.25	N/A	N/A	+0.33	N/A	N/A	N/A	N/A	+123	+97	

Nomenclature: See separate page